

3.1.2 Homogeneous Equations

A linear n th-order differential equation of the form

Note $y = 0$ is always a solution of a homogeneous linear equation.

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0 \quad (6)$$

is said to be **homogeneous**, whereas an equation

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x) \quad (7)$$

with $g(x)$ not identically zero is said to be **nonhomogeneous**. For example, $2y'' + 3y' - 5y = 0$ is a homogeneous linear second-order differential equation, whereas $x^2 y''' + 6y' + 10y = e^x$ is a nonhomogeneous linear third-order differential equation. The word *homogeneous* in this context does not refer to coefficients that are homogeneous functions as in Section 2.5; rather, the word has exactly the same meaning as in Section 2.3.

We shall see that in order to solve a nonhomogeneous linear equation (7), we must first be able to solve the **associated homogeneous equation** (6).

To avoid needless repetition throughout the remainder of this section, we shall, as a matter of course, make the following important assumptions when stating definitions and theorems about the linear equations (6) and (7). On some common interval I ,

Remember these assumptions in the definitions and theorems of this chapter.

- the coefficients $a_i(x)$, $i = 0, 1, 2, \dots, n$, are continuous;
- the right-hand member $g(x)$ is continuous; and
- $a_n(x) \neq 0$ for every x in the interval.

Differential Operators In calculus, differentiation is often denoted by the capital letter D ; that is, $dy/dx = Dy$. The symbol D is called a **differential operator** because it transforms a differentiable function into another function. For example, $D(\cos 4x) = -4 \sin 4x$, and $D(5x^3 - 6x^2) = 15x^2 - 12x$. Higher-order derivatives can be expressed in terms of D in a natural manner:

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2 y}{dx^2} = D(Dy) = D^2 y \quad \text{and in general} \quad \frac{d^n y}{dx^n} = D^n y,$$

where y represents a sufficiently differentiable function. Polynomial expressions involving D , such as $D + 3$, $D^2 + 3D - 4$, and $5x^3 D^3 - 6x^2 D^2 + 4xD + 9$, are also differential operators. In general, we define an **n th-order differential operator** to be

$$L = a_n(x)D^n + a_{n-1}(x)D^{n-1} + \cdots + a_1(x)D + a_0(x). \quad (8)$$

As a consequence of two basic properties of differentiation, $D(cf(x)) = c Df(x)$, c a constant, and $D\{f(x) + g(x)\} = Df(x) + Dg(x)$, the differential operator L possesses a **linearity property**; that is, L operating on a linear combination of two differentiable functions is the same as the linear combination of L operating on the individual functions. In symbols, this means

$$L\{\alpha f(x) + \beta g(x)\} = \alpha L(f(x)) + \beta L(g(x)), \quad (9)$$

where α and β are constants. Because of (9) we say that the n th-order differential operator L is a **linear operator**.

Differential Equations Any linear differential equation can be expressed in terms of the D notation. For example, the differential equation $y'' + 5y' + 6y = 5x - 3$ can be written as

$D^2 y + 5Dy + 6y = 5x - 3$ or $(D^2 + 5D + 6)y = 5x - 3$. Using (8), the linear n th-order differential equations (6) and (7) can be written compactly as

$$L(y) = 0 \quad \text{and} \quad L(y) = g(x),$$

Theorem 3.1.2 Superposition Principle—Homogeneous Equations

Let $y_1, y_2, \dots, y_k$ be solutions of the homogeneous n th-order differential equation (6) on an interval I . Then the linear combination

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_k y_k(x),$$

where the $c_i, i = 1, 2, \dots, k$ are arbitrary constants, is also a solution on the interval.

PROOF: We prove the case $k = 2$. Let L be the differential operator defined in (8), and let $y_1(x)$ and $y_2(x)$ be solutions of the homogeneous equation $L(y) = 0$. If we define $y = c_1 y_1(x) + c_2 y_2(x)$, then by linearity of L we have

$$L(y) = L\{c_1 y_1(x) + c_2 y_2(x)\} = c_1 L(y_1) + c_2 L(y_2) = c_1 \cdot 0 + c_2 \cdot 0 = 0. \quad \equiv$$

Corollaries to Theorem 3.1.2

- (a) A constant multiple $y = c_1 y_1(x)$ of a solution $y_1(x)$ of a homogeneous linear differential equation is also a solution.
- (b) A homogeneous linear differential equation always possesses the trivial solution $y = 0$.

Linear Dependence and Linear Independence The next two concepts are basic to the study of linear differential equations.

Definition 3.1.1 Linear Dependence/Independence

A set of functions $f_1(x), f_2(x), \dots, f_n(x)$ is said to be **linearly dependent** on an interval I if there exist constants $c_1, c_2, \dots, c_n$, not all zero, such that

$$c_1 f_1(x) + c_2 f_2(x) + \cdots + c_n f_n(x) = 0$$

for every x in the interval. If the set of functions is not linearly dependent on the interval, it is said to be **linearly independent**.

$c_1 \neq 0, \text{ or } c_2 \neq 0 \cdots c_n \neq 0$ (lin. dep.)
 $c_1 = c_2 = \cdots = c_n = 0$ (lin. indep.)

EXAMPLE 5 Linearly Dependent Functions

The functions $f_1(x) = \cos^2 x, f_2(x) = \sin^2 x, f_3(x) = \sec^2 x, f_4(x) = \tan^2 x$ are linearly dependent on the interval $(-\pi/2, \pi/2)$ since

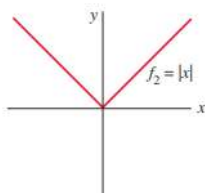
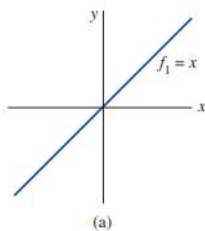
$$c_1 \cos^2 x + c_2 \sin^2 x + c_3 \sec^2 x + c_4 \tan^2 x = 0,$$

when $c_1 = c_2 = 1, c_3 = -1, c_4 = 1$. We used here $\cos^2 x + \sin^2 x = 1$ and $1 + \tan^2 x = \sec^2 x$ for every number x in the interval. $\equiv$

A set of n functions $f_1(x), f_2(x), \dots, f_n(x)$ is linearly dependent on an interval I if at least one of the functions can be expressed as a linear combination of the remaining functions. For example, three functions $f_1(x), f_2(x)$, and $f_3(x)$ are linearly dependent on I if at least one of these functions is a linear combination of the other two, say,

$$f_3(x) = c_1 f_1(x) + c_2 f_2(x)$$

for all x in I . A set of n functions is linearly independent on I if no one function is a linear combination of the other functions.



In other words, a set of functions is linearly independent on an interval if the only constants for which

$$c_1 f_1(x) + c_2 f_2(x) + \cdots + c_n f_n(x) = 0$$

for every x in the interval are $c_1 = c_2 = \cdots = c_n = 0$.

It is easy to understand these definitions in the case of two functions $f_1(x)$ and $f_2(x)$. If the functions are linearly dependent on an interval, then there exist constants c_1 and c_2 that are not both zero such that for every x in the interval $c_1 f_1(x) + c_2 f_2(x) = 0$. Therefore, if we assume that $c_1 \neq 0$, it follows that $f_1(x) = (-c_2/c_1)f_2(x)$; that is

If two functions are linearly dependent, then one is simply a constant multiple of the other.

Conversely, if $f_1(x) = c_2 f_2(x)$ for some constant c_2 , then $(-1) \cdot f_1(x) + c_2 f_2(x) = 0$ for every x on some interval. Hence the functions are linearly dependent, since at least one of the constants (namely, $c_1 = -1$) is not zero. We conclude that:

Two functions are linearly independent when neither is a constant multiple of the other on an interval.

Definition 3.1.2 Wronskian

Suppose each of the functions $f_1(x), f_2(x), \dots, f_n(x)$ possesses at least $n - 1$ derivatives. The determinant

$$W(f_1, f_2, \dots, f_n) = \begin{vmatrix} f_1 & f_2 & \cdots & f_n \\ f_1' & f_2' & \cdots & f_n' \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \cdots & f_n^{(n-1)} \end{vmatrix},$$

where the primes denote derivatives, is called the **Wronskian** of the functions.

The Wronskian determinant is named after the Polish philosopher and mathematician **Józef Maria Hoëné-Wronski** (1778–1853).

Theorem 3.1.3 Criterion for Linearly Independent Solutions

Let $y_1, y_2, \dots, y_n$ be n solutions of the homogeneous linear n th-order differential equation (6) on an interval I . Then the set of solutions is **linearly independent** on I if and only if $W(y_1, y_2, \dots, y_n) \neq 0$ for every x in the interval.

3.1.2 Homogeneous Equations

In Problems 15–22, determine whether the given set of functions is linearly dependent or linearly independent on the interval $(-\infty, \infty)$.

15. $f_1(x) = x, f_2(x) = x^2, f_3(x) = 4x - 3x^2$
16. $f_1(x) = 0, f_2(x) = x, f_3(x) = e^x$
17. $f_1(x) = 5, f_2(x) = \cos^2 x, f_3(x) = \sin^2 x$
18. $f_1(x) = \cos 2x, f_2(x) = 1, f_3(x) = \cos^2 x$
19. $f_1(x) = x, f_2(x) = x - 1, f_3(x) = x + 3$
20. $f_1(x) = 2 + x, f_2(x) = 2 + |x|$
21. $f_1(x) = 1 + x, f_2(x) = x, f_3(x) = x^2$
22. $f_1(x) = e^x, f_2(x) = e^{-x}, f_3(x) = \sinh x$

$$W(f_1, f_2) = \begin{vmatrix} 2+x & 2+|x| \\ 0+1 & 0+? \end{vmatrix}$$

we can't express $(2+x)$ in terms of $2+|x|$.
 $|x| \rightarrow$ distance, length
 $\text{diff}(\dots) = 0$

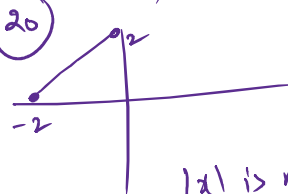
$W(f_1, f_2, f_3) \neq 0$ lin. indep.

$$W(f_1, f_2, f_3) = \begin{vmatrix} f_1 & f_2 & f_3 \\ f_1' & f_2' & f_3' \\ f_1'' & f_2'' & f_3'' \end{vmatrix} = \begin{vmatrix} 5 & \cos^2 x & \sin^2 x \\ 0 & 2\cos x(-\sin x) & 2\sin x \cos x \\ 0 & -2\cos x \sin x & 2\sin x \cos x \end{vmatrix}$$

$$= 5 \left[-2\sin 2x \cos 2x + 2\sin 2x \cos 2x \right] = 0$$

$\Rightarrow$ lin. dept.

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$|x|$ is not diff @ $x=0$
 but continuous at $x=0$

Conclusion: $W(f_1, f_2) = \begin{vmatrix} f_1 & f_2 \\ f_1' & f_2' \end{vmatrix} = 0 \Rightarrow$ lin. dept.

@ $x=0$, $f_1(x) = 2+0 = 2$ $f_1'(x) = 0$ $W(f_1, f_2) = 0$
 $f_2(x) = 2+1 \cdot 0 = 2$ $f_2'(x) = 0$ linearly dep.

verify that the given functions form a fundamental set of solutions of the differential equation on the indicated interval. Form the general solution.

23. $y'' - y' - 12y = 0$; $e^{-3x}, e^{4x}, (-\infty, \infty)$

24. $y'' - 4y = 0$; $\cosh 2x, \sinh 2x, (-\infty, \infty)$

25. $y'' - 2y' + 5y = 0$; $e^x \cos 2x, e^x \sin 2x, (-\infty, \infty)$

26. $4y'' - 4y' + y = 0$; $e^{x/2}, xe^{x/2}, (-\infty, \infty)$

27. $x^2 y'' - 6xy' + 12y = 0$; $x^3, x^4, (0, \infty)$

28. $x^2 y'' + xy' + y = 0$; $\cos(\ln x), \sin(\ln x), (0, \infty)$

29. $x^3 y''' + 6x^2 y'' + 4xy' - 4y = 0$; $x, x^{-2}, x^{-2} \ln x, (0, \infty)$

30. $y^{(4)} + y'' = 0$; $1, x, \cos x, \sin x, (-\infty, \infty)$

SA: (23) $W(f_1, f_2) = \begin{vmatrix} e^{-3x} & e^{4x} \\ -3e^{-3x} & 4e^{4x} \end{vmatrix}$
 $= 4e^{4x} - 3e^{-3x} \cdot e^{4x} = 4e^{4x} - 3e^{4x} = e^{4x} \neq 0$
 $y(x) = c_1 e^{-3x} + c_2 e^{4x}$

SA: (25) $W(f_1, f_2) = \begin{vmatrix} e^x \cos 2x & e^x \sin 2x \\ e^x (-\sin 2x) \cdot 2 + \cos 2x \cdot e^x & e^x \cos 2x \cdot 2 + \sin 2x \cdot e^x \end{vmatrix}$
 $= e^x \{ 2 \cos^2 2x + \sin^2 2x - 2 \sin^2 2x - \sin^2 2x \}$
 $= 2e^x$
 $y(x) = c_1 e^x \cos 2x + c_2 e^x \sin 2x$

Applications :- (Wronskian)

not direct mathematical application, but

2 vibration modes of vehicle are Independent

$W(y_1, y_2)$ are Independent (to verify)

↙ bounce ↘ Pitch

If $W=0$, It means Suspension Geometry Couples them, hitting a bump always causes both bounce & pitch we can't tune separately.

If $W \neq 0$, you can damp bounce without affecting pitch,

system is stable

This is why double bore strut matters.

$W(y_1, y_2)$ natural frequencies are Independent.

↙ Engine ↘ clutch hub

Hybrid → Dual mass Fly wheel

3.1.3 Nonhomogeneous Equations

In Problems 31–34, verify that the given two-parameter family of functions is the general solution of the nonhomogeneous differential equation on the indicated interval.

31. $y'' - 7y' + 10y = 24e^x$;

$$y = c_1 e^{2x} + c_2 e^{5x} + 6e^x, (-\infty, \infty)$$

32. $y'' + y = \sec x$;

$$y = c_1 \cos x + c_2 \sin x + x \sin x + (\cos x) \ln(\cos x), (-\pi/2, \pi/2)$$

33. $y'' - 4y' + 4y = 2e^{2x} + 4x - 12$;

$$y = c_1 e^{2x} + c_2 x e^{2x} + x^2 e^{2x} + x - 2, (-\infty, \infty)$$

34. $2x^2 y'' + 5xy' + y = x^2 - x$;

$$y = c_1 x^{-1/2} + c_2 x^{-1} + \frac{1}{15} x^2 - \frac{1}{6} x, (0, \infty)$$

35. (a) Verify that $y_{p_1} = 3e^{2x}$ and $y_{p_2} = x^2 + 3x$ are, respectively, particular solutions of

$$y'' - 6y' + 5y = -9e^{2x}$$

and $y'' - 6y' + 5y = 5x^2 + 3x - 16$.

(b) Use part (a) to find particular solutions of

$$y'' - 6y' + 5y = 5x^2 + 3x - 16 - 9e^{2x}$$

and $y'' - 6y' + 5y = -10x^2 - 6x + 32 + e^{2x}$.

36. (a) By inspection, find a particular solution of

$$y'' + 2y = 10.$$

(b) By inspection, find a particular solution of

$$y'' + 2y = -4x.$$

(c) Find a particular solution of $y'' + 2y = -4x + 10$.

(d) Find a particular solution of $y'' + 2y = 8x + 5$.

Sol (32) $y'' + y = \sec x$
 $\hookrightarrow \neq 0$

$$y(x) = c_1 \cos x + c_2 \sin x + x \sin x + (\cos x) \log(\cos x)$$

non homo DE

Gen solution
①

Particular
Solution
②

$$\textcircled{1} y'' + y = 0$$

$$W(y_1, y_2) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1 \neq 0$$

$$\therefore y(x) = c_1 \cos x + c_2 \sin x$$

$\downarrow$ $y_1(x)$ $\downarrow$ $y_2(x)$

$$\textcircled{2} y'' + y = 0$$

$$\left. \begin{array}{l} y_1(x) = \cos x \\ y_1'(x) = -\sin x \\ y_1''(x) = -\cos x \end{array} \right\} \text{Sub in}$$

$$-\cos x + \cos x = 0$$

$$\left. \begin{array}{l} y_2(x) = \sin x \\ y_2'(x) = \cos x \\ y_2''(x) = -\sin x \end{array} \right\} -\sin x + \sin x = 0$$

$\cos x, \sin x$ forms a fundamental solution, $W(y_1, y_2) \neq 0$
 $\Rightarrow$ lin Independent

3.3 Homogeneous Linear Equations with Constant Coefficients

INTRODUCTION We have seen that the linear first-order DE $y' + ay = 0$, where a is a constant, possesses the exponential solution $y = c_1 e^{-ax}$ on the interval $(-\infty, \infty)$. Therefore, it is natural to ask whether exponential solutions exist for homogeneous linear higher-order DEs

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0, \quad (1)$$

where the coefficients a_i , $i = 0, 1, \dots, n$ are real constants and $a_n \neq 0$. The surprising fact is that *all* solutions of these higher-order equations are either exponential functions or are constructed out of exponential functions.

Auxiliary Equation We begin by considering the special case of a second-order equation

$$ay'' + by' + cy = 0. \quad (2)$$

If we try a solution of the form $y = e^{mx}$, then after substituting $y' = me^{mx}$ and $y'' = m^2 e^{mx}$ equation (2) becomes

$$am^2 e^{mx} + bme^{mx} + ce^{mx} = 0 \quad \text{or} \quad e^{mx}(am^2 + bm + c) = 0.$$

Since e^{mx} is never zero for real values of x , it is apparent that the only way that this exponential function can satisfy the differential equation (2) is to choose m as a root of the quadratic equation

$$am^2 + bm + c = 0. \quad (3)$$

This last equation is called the **auxiliary equation** of the differential equation (2). Since the two roots of (3) are $m_1 = (-b + \sqrt{b^2 - 4ac})/2a$ and $m_2 = (-b - \sqrt{b^2 - 4ac})/2a$, there will be three forms of the general solution of (1) corresponding to the three cases:

- m_1 and m_2 are real and distinct ($b^2 - 4ac > 0$),
- m_1 and m_2 are real and equal ($b^2 - 4ac = 0$), and
- m_1 and m_2 are conjugate complex numbers ($b^2 - 4ac < 0$).

Case I: Distinct Real Roots Under the assumption that the auxiliary equation (3) has two unequal real roots m_1 and m_2 , we find two solutions, $y_1 = e^{m_1 x}$ and $y_2 = e^{m_2 x}$, respectively. We see that these functions are linearly independent on $(-\infty, \infty)$ and hence form a fundamental set. It follows that the general solution of (2) on this interval is

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}. \quad (4)$$

Case II: Repeated Real Roots When $m_1 = m_2$ we necessarily obtain only one exponential solution, $y_1 = e^{m_1 x}$. From the quadratic formula we find that $m_1 = -b/2a$ since the only way to have $m_1 = m_2$ is to have $b^2 - 4ac = 0$. It follows from the discussion in Section 3.2 that a second solution of the equation is

$$y_2 = e^{m_1 x} \int \frac{e^{2m_1 x}}{e^{2m_1 x}} dx = e^{m_1 x} \int dx = x e^{m_1 x}. \quad (5)$$

In (5) we have used the fact that $-b/a = 2m_1$. The general solution is then

$$y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}. \quad (6)$$

Case III: Conjugate Complex Roots If m_1 and m_2 are complex, then we can write $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$, where α and $\beta > 0$ are real and $i^2 = -1$. Formally, there is no difference between this case and Case I, hence

$$y = C_1 e^{(\alpha + i\beta)x} + C_2 e^{(\alpha - i\beta)x}$$

$\begin{matrix} \alpha + i\beta \\ \downarrow \\ \text{real part} \end{matrix} \rightarrow \begin{matrix} \text{imaginary part} \end{matrix}$

$$= e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$$

3.3 find the general solution of the given second-order differential equation.

1. $4y'' + y' = 0$
3. $y'' - y' - 6y = 0$
5. $y'' + 8y' + 16y = 0$
7. $12y'' - 5y' - 2y = 0$
9. $y'' + 9y = 0$
11. $y'' - 4y' + 5y = 0$
13. $3y'' + 2y' + y = 0$

Sol: ① $4m^2 + m = 0$

$$m(4m + 1) = 0$$

$$m = 0, m = -\frac{1}{4}$$

$$y(x) = C_1 e^{0x} + C_2 e^{-\frac{1}{4}x}$$

Sol: ③ $12m^2 - 5m - 2 = 0$

$$12m^2 - 8m + 3m - 2 = 0$$

$$4m(3m - 2) + 1(3m - 2) = 0$$

$$(4m + 1)(3m - 2) = 0$$

$$m = -\frac{1}{4}, m = \frac{2}{3}$$

$$y(x) = C_1 e^{-\frac{1}{4}x} + C_2 e^{\frac{2}{3}x}$$

25. $16 \frac{d^4 y}{dx^4} + 24 \frac{d^2 y}{dx^2} + 9y = 0$

26. $\frac{d^4 y}{dx^4} - 7 \frac{d^2 y}{dx^2} - 18y = 0$

27. $\frac{d^5 u}{dr^5} + 5 \frac{d^4 u}{dr^4} - 2 \frac{d^3 u}{dr^3} - 10 \frac{d^2 u}{dr^2} + \frac{du}{dr} + 5u = 0$

28. $2 \frac{d^5 x}{ds^5} - 7 \frac{d^4 x}{ds^4} + 12 \frac{d^3 x}{ds^3} + 8 \frac{d^2 x}{ds^2} = 0$

Sol: ②5 $16m^4 + 24m^2 + 9 = 0$

$$m^2 = t, 16(m^2)^2 + 24m^2 + 9 = 0$$

$$16t^2 + 24t + 9 = 0$$

$$16t^2 + 12t + 12t + 9 = 0$$

$$4t(4t + 3) + 3(4t + 3) = 0$$

$$(4t + 3)(4t + 3) = 0$$

$$t = -\frac{3}{4}, -\frac{3}{4}$$

$$m^2 = -\frac{3}{4}$$

$$m = \pm \sqrt{-\frac{3}{4}}$$

$$m = \pm \frac{i\sqrt{3}}{2} = 0 \pm \frac{i\sqrt{3}}{2}$$

$$y(x) = e^{0x} \left(C_1 \cos \frac{\sqrt{3}}{2} x + C_2 \sin \frac{\sqrt{3}}{2} x + C_3 \cos \frac{\sqrt{3}}{2} x + C_4 \sin \frac{\sqrt{3}}{2} x \right)$$

37. $y'' - 10y' + 25y = 0, y(0) = 1, y(1) = 0$

$$m^2 - 10m + 25 = 0$$

$$(m - 5)^2 = 0$$

$$m = 5, 5$$

$$y(x) = C_1 + C_2 \cdot x e^{5x}$$

$$y(0) = 1$$

$$1 = C_1 e^0 + C_2 e^0 \cdot 0$$

$$C_1 = 1$$

$$y(1) = 0$$

$$0 = C_1 e^5 + C_2 \cdot 1 e^5$$

$$0 = 1 \cdot e^5 + C_2 e^5$$

$$(1 + C_2) e^5 = 0$$

$$e^5 \neq 0 \Rightarrow 1 + C_2 = 0$$

$$C_2 = -1$$

$$\therefore y(x) = 1 - 1 \cdot x e^{5x}$$

Section: 3.4

In Problems 1-26, solve the given differential equation by undetermined coefficients.

1. $y'' + 3y' + 2y = 6$
2. $4y'' + 9y = 15$
3. $y'' - 10y' + 25y = 30x + 3$
4. $y'' + y' - 6y = 2x$
5. $\frac{1}{4}y'' + y' + y = x^2 - 2x$
6. $y'' - 8y' + 20y = 100x^2 - 26xe^x$
7. $y'' + 3y = -48x^2e^{3x}$
8. $4y'' - 4y' - 3y = \cos 2x$
9. $y'' - y' = -3$
10. $y'' + 2y' = 2x + 5 - e^{-2x}$
11. $y'' - y' + \frac{1}{4}y = 3 + e^{x/2}$
12. $y'' - 16y = 2e^{4x}$
13. $y'' + 4y = 3 \sin 2x$
14. $y'' - 4y = (x^2 - 3) \sin 2x$
15. $y'' + y = 2x \sin x$
16. $y'' - 5y' = 2x^3 - 4x^2 - x + 6$
17. $y'' - 2y' + 5y = e^x \cos 2x$
18. $y'' - 2y' + 2y = e^{2x}(\cos x - 3 \sin x)$
19. $y'' + 2y' + y = \sin x + 3 \cos x$
20. $y'' + 2y' - 24y = 16 - (x + 2)e^{4x}$
21. $y''' - 6y'' = 3 - \cos x$
22. $y''' - 2y'' - 4y' + 8y = 6xe^{2x}$
23. $y''' - 3y'' + 3y' - y = x - 4e^x$
24. $y''' - y'' - 4y' + 4y = 5 - e^x + e^{2x}$

If RHS is linear
" " " and "
" " " and "

Particular solution

$$y_p(x) = Ax + B$$

$$y_p(x) = Ax^2 + Bx + C$$

Sol: 3 $m^2 - 10m + 25 = 0$
($m - 5$)² = 0
 $m = 5, 5$

Comp. fun $y(x) = C_1 e^{5x} + C_2 x \cdot e^{5x}$
(CF)

$g(x) = 30x + 3$ (linear)

$y_p(x) = Ax + B$

$y_p'(x) = A \cdot 1 + 0$

$y_p''(x) = 0$

Sub in DE

$y_p'' - 10y_p' + 25y_p = 30x + 3$

$0 - 10A + 25(Ax + B) = 30x + 3$

Equating x on b.s., $25A = 30$ $A = 6/5$

" Const " " , $-10A + 25B = 3$

$-10(6/5) + 25B = 3$

$B = 3/5$

$y_p(x) = \frac{6}{5}x + \frac{3}{5}$ PI

G.S. = CF + PI

6) $y'' - 8y' + 20y = 100x^2 - 26xe^x$

AE $m^2 - 8m + 20 = 0$

$m = 4 \pm 2i$

$y(x) = e^{4x} (C_1 \cos 2x + C_2 \sin 2x)$ CF

$g(x) = 100x^2 - 26xe^x$

2nd degree
1st degree

$y_p(x) = (Ax^2 + Bx + C) + (Dx + E)e^x$

$y_p'(x) = 2Ax + B + D \cdot e^x + (Dx + E)e^x$

$y_p''(x) = 2A + D \cdot e^x + (Dx + E)e^x + e^x \cdot D$

$y_p'''(x) = 2A + 2D \cdot e^x + (Dx + E)e^x$

$y_p'' - 8y_p' + 20y_p = 100x^2 - 26xe^x$

$2A + 2D \cdot e^x + (Dx + E)e^x - 8(2Ax + B + D \cdot e^x + (Dx + E)e^x) + 20(Ax^2 + Bx + C) + (Dx + E)e^x = 100x^2 - 26xe^x$

Equating x^2 term, $20A = 100$ $A = 5$

" xe^x " , $D - 8D + 20D = -26$ $D = -2$

" x term " , $-16A + 20B = 0$ $B = -2$

" Constant " , $2A - 8B + 20C = 0$ $C = \frac{6}{20}$

Sub $x = 0$, $E = 2$

$y_p(x) = (5x^2 + 2x + \frac{6}{20}) + (-2x + 2)e^x$ PI

G.S. = CF + PI

33. $\frac{d^2x}{dt^2} + \omega^2 x = F_0 \sin \omega t, x(0) = 0, x'(0) = 0$

If RHS $\sin \theta$ or $\cos \theta$

34. $\frac{d^2x}{dt^2} + \omega^2 x = F_0 \cos \gamma t, x(0) = 0, x'(0) = 0$

$y_p(x) = A \cos \alpha x + B \sin \alpha x$

35. $y''' - 2y'' + y' = 2 - 24e^x + 40e^{5x}, y(0) = \frac{1}{2}, y'(0) = \frac{5}{2}, y''(0) = -\frac{9}{2}$

sol: 34 $m^2 + \omega^2 = 0$
 $m^2 = -\omega^2$
 $m = \pm i\omega$

36. $y''' + 8y = 2x - 5 + 8e^{-2x}, y(0) = -5, y'(0) = 3, y''(0) = -4$

$\alpha = 0 \quad \beta = \omega$

$x(t) = e^{0t} (C_1 \cos \omega t + C_2 \sin \omega t)$ CF

PI $x_p(t) = A \cos \gamma t + B \sin \gamma t$
 $x_p'(t) = -A \sin \gamma t (\gamma) + B \cos \gamma t (\gamma)$
 $x_p''(t) = -A \gamma^2 \cos \gamma t - B \gamma^2 \sin \gamma t$ Sub in DE

$x_p''(t) + \omega^2 x_p(t) = F_0 \cos \gamma t$

$-A \gamma^2 \cos \gamma t - B \gamma^2 \sin \gamma t + \omega^2 (A \cos \gamma t + B \sin \gamma t) = F_0 \cos \gamma t$

Equating $\cos \gamma t$, $-A \gamma^2 + A \omega^2 = F_0$
 $A(\omega^2 - \gamma^2) = F_0 \Rightarrow A = \frac{F_0}{\omega^2 - \gamma^2}$

Equating $\sin \gamma t$, $-B \gamma^2 + B \omega^2 = 0$
 $B(\omega^2 - \gamma^2) = 0$

$B = 0$ $-\gamma^2 + \omega^2 \neq 0$

$x_p(t) = \frac{F_0}{\omega^2 - \gamma^2} \cos \gamma t$ PI

GS = CF + PI

$x(t) = (C_1 \cos \omega t + C_2 \sin \omega t) + \frac{F_0}{\omega^2 - \gamma^2} \cos \gamma t$

$t \uparrow$
 $x(0) = 0$

$0 = C_1 \cos 0 + C_2 \sin 0 + \frac{F_0}{\omega^2 - \gamma^2} \cos 0$

$0 = C_1 + \frac{F_0}{\omega^2 - \gamma^2}$

$C_1 = \frac{F_0}{\gamma^2 - \omega^2}$

$x'(0) = 0$

$x'(t) = -C_1 \sin \omega t + C_2 \cos \omega t + \frac{F_0 \sin \gamma t}{\gamma^2 - \omega^2}$

$x'(0) = 0$

$0 = -C_1 \sin 0 + C_2 \cos 0 + \frac{F_0 \sin 0}{\gamma^2 - \omega^2}$

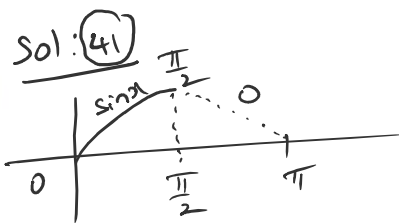
$C_2 = 0$

GS $x(t) = \left(\frac{F_0}{\gamma^2 - \omega^2} \cos \omega t + 0 \right) + \frac{F_0}{\omega^2 - \gamma^2} \cos \gamma t$

x

Section 3.4

In Problems 41 and 42, solve the given initial-value problem in which the input function $g(x)$ is discontinuous. [Hint: Solve each problem on two intervals, and then find a solution so that y and y' are continuous at $x = \pi/2$ (Problem 41) and at $x = \pi$ (Problem 42).]



41. $y'' + 4y = g(x)$, $y(0) = 1$, $y'(\pi/2) = 2$, where

$$g(x) = \begin{cases} \sin x, & 0 \leq x \leq \pi/2 \\ 0, & x > \pi/2 \end{cases}$$

42. $y'' - 2y' + 10y = g(x)$, $y(0) = 0$, $y'(\pi) = 0$, where

$$g(x) = \begin{cases} 20, & 0 \leq x \leq \pi \\ 0, & x > \pi \end{cases}$$

AE $m^2 + 4 = 0 \Rightarrow m^2 = -4$

$m = \pm 2i$

$y_c(x) = C_1 \cos 2x + C_2 \sin 2x$

$g(x) = \sin x$

$y_p = A \cos x + B \sin x \rightarrow (1)$

$y'_p = -A \sin x + B \cos x$

$y''_p = -A \cos x - B \sin x \rightarrow (2)$

Subst (1), (2) in (I) $\leftarrow$
 $-A \cos x - B \sin x + 4(A \cos x + B \sin x) = \sin x$

Equating $\cos x$, $3A = 0 \Rightarrow A = 0$
 " $\sin x$, $3B = 1 \Rightarrow B = \frac{1}{3}$ Subst in (1)

(1) $\Rightarrow y_p = \frac{1}{3} \sin x$

Gen. solution: $y(x) = C_1 \cos 2x + C_2 \sin 2x + \frac{1}{3} \sin x$

For $g(x) = \sin x$
 $y'(x) = -C_1 \sin 2x (2) + C_2 \cos 2x (2) + \frac{1}{3} \cos x$

$y(0) = 1$

$1 = C_1 \cos 0 + C_2 \sin 0 + \frac{1}{3} \sin 0$

$C_1 = 1$

$y'(\pi/2) = 2$

$2 = -C_1 \sin 0 + C_2 \cos 0 (2) + \frac{1}{3} \cos 0$

$2 = 2C_2 + \frac{1}{3}$

$C_2 = \frac{5}{6}$

$\therefore y(x) = 1 \cdot \cos 2x + \frac{5}{6} \sin 2x + \frac{1}{3} \sin x$

$0 \leq x \leq \frac{\pi}{2}$

For $g(x) = 0$, $x > \frac{\pi}{2}$

Gen. solution: $y(x) = C_3 \cos 2x + C_4 \sin 2x + 0 \rightarrow (4)$

From (3), $y(x) = \cos 2x + \frac{5}{6} \sin 2x + \frac{1}{3} \sin x$

$$② \quad x = \frac{\pi}{2} \quad y\left(\frac{\pi}{2}\right) = \cos 2\left(\frac{\pi}{2}\right) + \frac{5}{6} \cdot \sin 2\left(\frac{\pi}{2}\right) + \frac{1}{3} \sin \frac{\pi}{2}$$

$$\begin{aligned} &= -1 + 0 + \frac{1}{3} \\ \text{B.C.'s} \quad &\boxed{y\left(\frac{\pi}{2}\right) = -\frac{2}{3}} \rightarrow (5) \end{aligned}$$

$$\text{Diff } (3), \quad y'(x) = -\sin 2x(2) + \frac{5}{6} \cos 2x(2) + \frac{1}{3} \cdot \cos x \cdot 1$$

$$\uparrow \text{ at } x = \frac{\pi}{2}$$

$$y'\left(\frac{\pi}{2}\right) = -2 \cdot \sin\left(2 \cdot \frac{\pi}{2}\right) + \frac{5}{3} \cdot \cos\left(2 \cdot \frac{\pi}{2}\right) + \frac{1}{3} \cdot \cos \frac{\pi}{2}$$

$$\boxed{y'\left(\frac{\pi}{2}\right) = \frac{5}{3}} \rightarrow (6)$$

∴ Find c_3, c_4 , we need to use (5), (6) (B.C.'s)

$$y\left(\frac{\pi}{2}\right) = c_3 \cdot \cos 2\left(\frac{\pi}{2}\right) + c_4 \sin 2\left(\frac{\pi}{2}\right) \quad \left| \quad y'(x) = c_3 \sin 2x(-2) + c_4 \cos 2x(2)\right.$$

$$-\frac{2}{3} = c_3(-1) + c_4(0)$$

$$\boxed{c_3 = 2/3}$$

$$y'\left(\frac{\pi}{2}\right) = -2c_3 \cdot \sin 2\frac{\pi}{2} + 2c_4 \cos 2\frac{\pi}{2}$$

$$\frac{5}{3} = 2c_4(-1)$$

$$\boxed{c_4 = -\frac{5}{6}}$$

$$\text{Sub in (4)} \quad \left. \begin{matrix} c_3, c_4 \end{matrix} \right) y(x) = \frac{2}{3} \cdot \cos 2x + \frac{5}{6} \cdot \sin 2x$$

3.6 Cauchy–Euler Equations

INTRODUCTION The relative ease with which we were able to find explicit solutions of linear higher-order differential equations with *constant coefficients* in the preceding sections does not, in general, carry over to linear equations with *variable coefficients*. We shall see in Chapter 5 that when a linear differential equation has variable coefficients, the best that we can usually expect is to find a solution in the form of an infinite series. However, the type of differential equation considered in this section is an exception to this rule; it is an equation with variable coefficients whose general solution can always be expressed in terms of powers of x , sines, cosines, logarithmic, and exponential functions. Moreover, its method of solution is quite similar to that for constant equations.

Cauchy–Euler Equation Any linear differential equation of the form

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 x \frac{dy}{dx} + a_0 y = g(x),$$

where the coefficients $a_n, a_{n-1}, \dots, a_0$ are constants, is known diversely as a **Cauchy–Euler equation**, an **Euler–Cauchy equation**, an **Euler equation**, or an **equidimensional equation**. The differential equation is named in honor of two of the most prolific mathematicians of all time, **Augustin-Louis Cauchy** (French, 1789–1857) and **Leonhard Euler** (Swiss, 1707–1783). The observable characteristic of this type of equation is that the degree $k = n, n-1, \dots, 1, 0$ of the monomial coefficients x^k matches the order k of differentiation $d^k y/dx^k$:

$$\begin{array}{ccc} \text{same} & & \text{same} \\ \downarrow & & \downarrow \\ a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots \end{array}$$

As in Section 3.3, we start the discussion with a detailed examination of the forms of the general solutions of the homogeneous second-order equation

$$ax^2 \frac{d^2 y}{dx^2} + bx \frac{dy}{dx} + cy = 0. \quad (1)$$

The solution of higher-order equations follows analogously. Also, we can solve the nonhomogeneous equation $ax^2 y'' + bxy' + cy = g(x)$ by variation of parameters, once we have determined the complementary function $y_c(x)$ of (1).

The coefficient of $d^2 y/dx^2$ is zero at $x = 0$. Hence, in order to guarantee that the fundamental results of Theorem 3.1.1 are applicable to the Cauchy–Euler equation, we confine our attention to finding the general solution on the interval $(0, \infty)$. Solutions on the interval $(-\infty, 0)$ can be obtained by substituting $t = -x$ into the differential equation. See Problems 49 and 50 in Exercises 3.6.

How to find the solution? Sub ② in ①

$$ax^2 \cdot y'' + bx y' + cy = 0 \rightarrow \textcircled{1} \quad \left. \begin{array}{l} \text{Let } y = x^m \\ y' = m x^{m-1} \\ y'' = m(m-1) x^{m-2} \end{array} \right\} \rightarrow \textcircled{2}$$

$$a x^2 m(m-1) x^{m-2} + b x \cdot m x^{m-1} + c \cdot x^m = 0$$

$$a m(m-1) x^m + b \cdot m x^m + c x^m = 0$$

Aux. Eqn $am^2 + (b-a)m + c = 0$

Case ① Distinct real roots $y_1 = x^{m_1}, y_2 = x^{m_2}$

(CF) $y_c = C_1 \cdot x^{m_1} + C_2 \cdot x^{m_2}$

Case ② Repeated roots $y_c = C_1 \cdot x^{m_1} + C_2 x^{m_2} \cdot \log x$

Case ② Complex roots $y = x^m (C_1 \cdot \cos(\beta \cdot \log x) + C_2 \cdot \sin(\beta \cdot \log x))$

Method of Solution We try a solution of the form $y = x^m$, where m is to be determined. Analogous to what happened when we substituted e^{mx} into a linear equation with constant coefficients, after substituting x^m each term of a Cauchy-Euler equation becomes a polynomial in m times x^m since

$$\begin{aligned} a_k x^k \frac{d^k y}{dx^k} &= a_k x^k m(m-1)(m-2) \cdots (m-k+1) x^{m-k} \\ &= a_k m(m-1)(m-2) \cdots (m-k+1) x^m. \end{aligned}$$

For example, by substituting $y = x^m$ the second-order equation (1) becomes

$$ax^2 \frac{d^2 y}{dx^2} + bx \frac{dy}{dx} + cy = am(m-1)x^m + bmx^m + cx^m = (am(m-1) + bm + c)x^m = 0.$$

Thus $y = x^m$ is a solution of the differential equation whenever m is a solution of the **auxiliary equation**

$$am(m-1) + bm + c = 0 \quad \text{or} \quad am^2 + (b-a)m + c = 0. \quad (2)$$

There are three different cases to be considered, depending on whether the roots of this quadratic equation are real and distinct, real and equal, or complex. In the last case the roots appear as a conjugate pair.

Case I: Distinct Real Roots Let m_1 and m_2 denote the real roots of (2) such that $m_1 \neq m_2$. Then $y_1 = x^{m_1}$ and $y_2 = x^{m_2}$ form a fundamental set of solutions. Hence the general solution of (1) is

$$y = c_1 x^{m_1} + c_2 x^{m_2}. \quad (3)$$

Problem 25 to 30 for IVP solve

② $x^2 y'' + 3xy' = 0 \quad y(1)=0, \quad y'(1)=4$

A.E $m^2 + 3m = 0$
 $m(m+3) = 0$
 $m = 0, -3$

$y_c(x) = C_1 \cdot e^{0x} + C_2 \cdot e^{-3x}$
 $y'_c(x) = C_2 \cdot e^{-3x} (-3)$

$y(1)=0$
 $y(1) = C_1 + C_2 e^{-3}$
 $0 = C_1 + C_2 e^{-3}$
 $0 = C_1 - \frac{4}{3} \cdot e^3 \cdot e^{-3}$

$y'(1)=4$
 $4 = -3C_2 \cdot e^{-3}$
 $\frac{-4}{3e^3} = C_2$ or $C_2 = -\frac{4}{3} e^3$

$C_1 = \frac{4}{3}$

$\therefore y_c(x) = \frac{4}{3} - \frac{4}{3} \cdot e^3 \cdot e^{-3x}$

In Problems 33–38, find a homogeneous Cauchy–Euler → degree of Polyn = order of DE differential equation whose general solution is given.

33. $y = c_1 x^4 + c_2 x^{-2}$

34. $y = c_1 + c_2 x^5$

35. $y = c_1 x^{-3} + c_2 x^{-3} \ln x$

36. $y = c_1 + c_2 x + c_3 x \ln x$

37. $y = c_1 \cos(\ln x) + c_2 \sin(\ln x)$

38. $y = c_1 x^{1/2} \cos(\frac{1}{2} \ln x) + c_2 x^{1/2} \sin(\frac{1}{2} \ln x)$

Sol 33:- Distinct roots

$$y = c_1 x^{m_1} + c_2 x^{m_2}$$

$$m_1 = 4 \quad m_2 = -2$$

$$A.E \quad (m-4)(m+2) = 0$$

$$m^2 - 2m - 8 = 0 \quad (am^2 + (b-a)m + c = 0)$$

DE $ax^2 y'' + bxy' + cy = 0$

Coeff m^2 $a^2 = 1 \Rightarrow a = 1$

$$b-a = -2$$

$$b-1 = -2 \Rightarrow b = -1$$

Constant $c = -8$

$$\rightarrow DE \quad 1 \cdot x^2 y'' - 1 \cdot xy' - 8y = 0.$$

Sol: 38

$$y = x^\alpha (c_1 \cos(\beta \cdot \log_e x) + c_2 \sin(\beta \cdot \log_e x))$$

$$\alpha = \frac{1}{2}, \beta = \frac{1}{2} \quad m = \alpha \pm i\beta$$

$$= \frac{1}{2} \pm i\frac{1}{2}$$

$$m = \left(\frac{1}{2} + i\frac{1}{2}\right) \left(\frac{1}{2} - i\frac{1}{2}\right)$$

$$\left(\underbrace{m - \frac{1}{2}}_a - \underbrace{\frac{1}{2}i}_b\right) \left(\underbrace{m - \frac{1}{2}}_a + \underbrace{\frac{1}{2}i}_b\right) = 0$$

$$(m - \frac{1}{2})^2 - (\frac{1}{2}i)^2 = 0 \quad (i^2 = -1)$$

$$m^2 - 2m \cdot \frac{1}{2} + \frac{1}{4} - \frac{1}{4}(-1) = 0$$

$$m^2 - m + \frac{1}{2} = 0$$

$$2m^2 - 2m + 1 = 0$$

$$am^2 + (b-a)m + c = 0$$

DE $ax^2 y'' + bxy' + cy = 0$

Giving m^2, m & Constant

$$a=2, \quad b-a = -2, \quad c=1$$

$$b=0$$

$$\therefore DE \quad 2x^2 y'' + y = 0.$$

In Problems 43–48, use the substitution $x = e^t$ to transform the given Cauchy–Euler equation to a differential equation with constant coefficients. Solve the original equation by solving the new equation using the procedures in Sections 3.3–3.5.

43. $x^2 y'' + 9xy' - 20y = 0$ 44. $x^2 y'' - 9xy' + 25y = 0$

45. $x^2 y'' + 10xy' + 8y = x^2$ 46. $x^2 y'' - 4xy' + 6y = \ln x^2$

Sol: 44 $x = e^t$

$$(e^t)^2 y'' - 9e^t y' + 25e^t = 0 \rightarrow (1)$$

To transform x in terms of t

$$x = e^t$$

$$\left(\frac{dx}{dt} = e^t \right) \rightarrow (*)$$

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

$$\frac{dy}{dt} = y'(x) \cdot e^t$$

$$x = e^t$$

$$\log_e x = \log_e e^t$$

$$\log_e x = t$$

$$y'(x) = \frac{1}{e^t} \cdot \frac{dy}{dt} \rightarrow (2)$$

$$y''(x) = \frac{d}{dx} \left(e^{-t} \cdot \frac{dy}{dt} \right)$$

$$= e^{-t} \frac{d}{dt} \left(\frac{dy}{dt} \right) \left(\frac{dt}{dx} \right) + \frac{dy}{dt} (e^{-t}) \left(\frac{dt}{dx} \right) \downarrow (*)$$

$$= e^{-t} \cdot \frac{d^2 y}{dt^2} e^{-t} + \frac{dy}{dt} \cdot e^{-t} \cdot e^{-t}$$

$$y'' = e^{-2t} \frac{d^2 y}{dt^2} + e^{-2t} \frac{dy}{dt} \rightarrow (3)$$

$$(2), (3) \text{ in } (1)$$

$$e^{2t} \left(e^{-2t} \frac{d^2 y}{dt^2} + e^{-2t} \frac{dy}{dt} \right) - 9e^t \left(e^{-t} \frac{dy}{dt} \right) + 25e^t = 0$$

$$\frac{d^2 y}{dt^2} + \frac{dy}{dt} - 9 \frac{dy}{dt} + 25y = 0$$

$$\frac{d^2 y}{dt^2} - 8 \frac{dy}{dt} + 25y = 0$$

$$\text{A.E. } m^2 - 10m + 25 = 0$$

$$(m-5)^2 = 0 \quad m = 5, 5$$

$$y(x) = c_1 e^{5t} + c_2 \cdot e^{5t} \cdot t$$

$$= c_1 e^{5 \log_e x} + c_2 e^{5 \log_e x} \cdot (\log_e x)$$

$$= c_1 e^{\log_e x^5} + c_2 \cdot e^{\log_e x^5} \cdot \log_e x$$

$$y(x) = c_1 \cdot x^5 + c_2 \cdot x^5 \cdot \log_e x$$

In Problems 49 and 50, use the substitution $t = -x$ to solve the given initial-value problem on the interval $(-\infty, 0)$.

49. $4x^2y'' + y = 0$, $y(-1) = 2$, $y'(-1) = 4$

50. $x^2y'' - 4xy' + 6y = 0$, $y(-2) = 8$, $y'(-2) = 0$

INTRODUCTION In this section we are going to consider several linear dynamical systems in which each mathematical model is a linear second-order differential equation with constant coefficients along with initial conditions specified at time t_0 :

$$a_2 \frac{d^2 y}{dt^2} + a_1 \frac{dy}{dt} + a_0 y = g(t), \quad y(t_0) = y_0, \quad y'(t_0) = y_1.$$

The function g is variously called the **input**, **driving**, or **forcing function** of the system. The **output** or **response** of the system is a function $y(t)$ defined on an I interval containing t_0 that satisfies both the differential equation and the initial conditions on the interval I .

damping
↓
amt of stretch
or elongation
↓
depend
on
mass
 m

$F \propto s$
↗ elongation
 $F = -ks$
 $|F| = |-ks|$

$$F = k|s|$$

$$\downarrow$$

 $10 \text{ lb} = k \cdot \frac{1}{2} \text{ ft}$

$$k = 20 \text{ lb/ft}$$

$$8 \text{ lb} = k \cdot \frac{2}{5} \text{ ft}$$

$$k = 20 \text{ lb/ft}$$

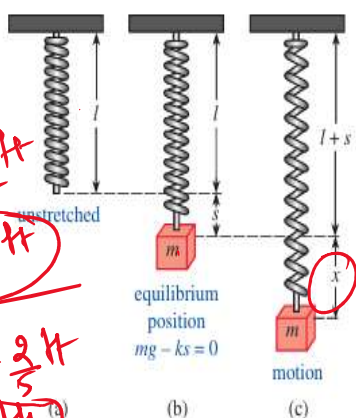


FIGURE 3.8.1 Spring/mass system

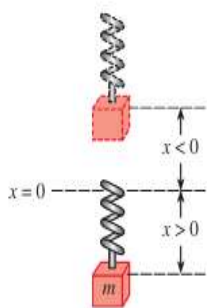


FIGURE 3.8.2 Positive direction is below equilibrium position

3.8.1 Spring/Mass Systems: Free Undamped Motion

Hooke's Law Suppose a flexible spring is suspended vertically from a rigid support and then a mass m is attached to its free end. The amount of stretch, or elongation, of the spring will, of course, depend on the mass; masses with different weights stretch the spring by differing amounts. By Hooke's law, the spring itself exerts a restoring force F opposite to the direction of elongation and **proportional to the amount of elongation s** . Simply stated, $F = -ks$, where $k > 0$ is a constant of proportionality called the **spring constant**. The spring is essentially characterized by the number k . Using $|F| = k|s|$, we see that if a mass weighing 10 lb stretches a spring $\frac{1}{2}$ ft, then $10 = k(\frac{1}{2})$ implies $k = 20 \text{ lb/ft}$. Necessarily then, a mass weighing, say, 8 lb stretches the same spring only $\frac{2}{5}$ ft.

Newton's Second Law After a mass m is attached to a spring, it stretches the spring by an amount s and attains a position of equilibrium at which its weight W is balanced by the restoring force ks . Recall that weight is defined by $W = mg$, where mass is measured in slugs, kilograms, or grams and $g = 32 \text{ ft/s}^2$, 9.8 m/s^2 , or 980 cm/s^2 , respectively. As indicated in FIGURE 3.8.1(b), the condition of equilibrium is $mg = ks$ or $mg - ks = 0$. If the mass is displaced by an amount x from its equilibrium position, the restoring force of the spring is then $k(x + s)$. Assuming that there are no retarding forces acting on the system and assuming that the mass vibrates free of other external forces—**free motion**—we can equate Newton's second law with the net, or resultant, force of the restoring force and the weight:

$$m \frac{d^2 x}{dt^2} = -k(s + x) + mg = -kx + \underbrace{mg - ks}_{\text{zero}} = -kx. \quad (1)$$

The negative sign in (1) indicates that the restoring force of the spring acts opposite to the direction of motion. Furthermore, we can adopt the convention that displacements measured below the equilibrium position are positive. See FIGURE 3.8.2.

$$F = ma$$

$$= m \cdot \frac{d^2 x}{dt^2} \rightarrow (2)$$

By Newton's second law
net restoring force = (mass $\times$ acceln)

$$-ks - kx + W = m \cdot \frac{d^2 x}{dt^2}$$

$$-ks - kx + mg = m \cdot \frac{d^2 x}{dt^2}$$

$$v = \frac{dx}{dt} \rightarrow \text{displ}$$

$$a = \frac{dv}{dt} \rightarrow \text{acceln}$$

$$a = \frac{d}{dt} \left(\frac{dx}{dt} \right)$$

↗ no external force

$$-kx + kx + kx = m \cdot \frac{d^2x}{dt^2}$$

$$m \cdot \frac{d^2x}{dt^2} = kx \Rightarrow \frac{d^2x}{dt^2} = \frac{k}{m}x$$

DE of Free Undamped Motion By dividing (1) by the mass m we obtain the second-order differential equation $d^2x/dt^2 + (k/m)x = 0$ or

$$\frac{d^2x}{dt^2} + \omega^2 x = 0,$$

$$\frac{d^2x}{dt^2} - \frac{k}{m}x = 0$$

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \quad (2)$$

where $\omega^2 = k/m$. Equation (2) is said to describe **simple harmonic motion** or **free undamped motion**. Two obvious initial conditions associated with (2) are $x(0) = x_0$, the amount of initial displacement, and $x'(0) = x_1$, the initial velocity of the mass. For example, if $x_0 > 0$, $x_1 < 0$, the mass starts from a point *below* the equilibrium position with an imparted *upward* velocity. When $x_1 = 0$ the mass is said to be released from *rest*. For example, if $x_0 < 0$, $x_1 = 0$, the mass is released from rest from a point $|x_0|$ units *above* the equilibrium position.

Solution and Equation of Motion To solve equation (2) we note that the solutions of the auxiliary equation $m^2 + \omega^2 = 0$ are the complex numbers $m_1 = \omega i$, $m_2 = -\omega i$. Thus from (8) of Section 3.3 we find the general solution of (2) to be

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t. \quad (3)$$

DE of Free Damped Motion In the study of mechanics, damping forces acting on a body are considered to be proportional to a power of the instantaneous velocity. In particular, we shall assume throughout the subsequent discussion that this force is given by a constant multiple of dx/dt . When no other external forces are impressed on the system, it follows from Newton's second law that

$$m \frac{d^2x}{dt^2} = -kx - \beta \frac{dx}{dt}, \quad (10)$$

where β is a positive *damping constant* and the negative sign is a consequence of the fact that the damping force acts in a direction opposite to the motion.

Dividing (10) by the mass m , we find the differential equation of **free damped motion** is $d^2x/dt^2 + (\beta/m)dx/dt + (k/m)x = 0$ or

$$\frac{d^2x}{dt^2} + 2\lambda \frac{dx}{dt} + \omega^2 x = 0, \quad (11)$$

where $2\lambda = \frac{\beta}{m}, \quad \omega^2 = \frac{k}{m}. \quad (12)$

The symbol 2λ is used only for algebraic convenience, since the auxiliary equation is $m^2 + 2\lambda m + \omega^2 = 0$ and the corresponding roots are then

$$m_1 = -\lambda + \sqrt{\lambda^2 - \omega^2}, \quad m_2 = -\lambda - \sqrt{\lambda^2 - \omega^2}.$$

We can now distinguish three possible cases depending on the algebraic sign of $\lambda^2 - \omega^2$. Since each solution contains the *damping factor* $e^{-\lambda t}$, $\lambda > 0$, the displacements of the mass become negligible over a long period of time.

k - Spring Constant = $\begin{cases} \text{higher, stiffer, sportier but bumpier} \\ \text{lower, softer, comfy but flopsy.} \end{cases}$

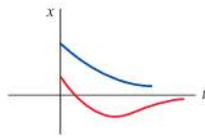


FIGURE 3.8.8 Motion of an overdamped system

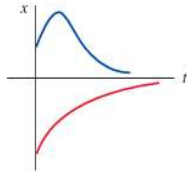
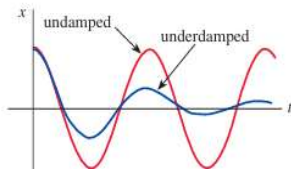


FIGURE 3.8.9 Motion of a critically damped system



We can now distinguish three possible cases depending on the algebraic sign of $\lambda^2 - \omega^2$. Since each solution contains the *damping factor* $e^{-\lambda t}$, $\lambda > 0$, the displacements of the mass become negligible over a long period of time.

Case I: $\lambda^2 - \omega^2 > 0$ In this situation the system is said to be **overdamped** because the damping coefficient β is large when compared to the spring constant k . The corresponding solution of (11) is $x(t) = c_1 e^{m_1 t} + c_2 e^{m_2 t}$ or

$$x(t) = e^{-\lambda t} (c_1 e^{\sqrt{\lambda^2 - \omega^2} t} + c_2 e^{-\sqrt{\lambda^2 - \omega^2} t}). \quad (13)$$

Equation 13 represents a smooth and nonoscillatory motion. FIGURE 3.8.8 shows two possible graphs of $x(t)$.

Case II: $\lambda^2 - \omega^2 = 0$ The system is said to be **critically damped** because any slight decrease in the damping force would result in oscillatory motion. The general solution of (11) is $x(t) = c_1 e^{m_1 t} + c_2 t e^{m_1 t}$ or

$$x(t) = e^{-\lambda t} (c_1 + c_2 t). \quad (14)$$

Some graphs of typical motion are given in FIGURE 3.8.9. Notice that the motion is quite similar to that of an overdamped system. It is also apparent from (14) that the mass can pass through the equilibrium position at most one time.

Case III: $\lambda^2 - \omega^2 < 0$ In this case the system is said to be **underdamped** because the damping coefficient is small compared to the spring constant. The roots m_1 and m_2 are now complex:

$$m_1 = -\lambda + \sqrt{\omega^2 - \lambda^2} i, \quad m_2 = -\lambda - \sqrt{\omega^2 - \lambda^2} i.$$

Thus the general solution of equation (11) is

$$x(t) = e^{-\lambda t} (c_1 \cos \sqrt{\omega^2 - \lambda^2} t + c_2 \sin \sqrt{\omega^2 - \lambda^2} t). \quad (15)$$

3.8.1 Spring/Mass Systems: Free Undamped Motion

1. A mass weighing 4 pounds is attached to a spring whose spring constant is 16 lb/ft. What is the period of simple harmonic motion?
2. A 20-kilogram mass is attached to a spring. If the frequency of simple harmonic motion is $2/\pi$ cycles/s, what is the spring constant k ? What is the frequency of simple harmonic motion if the original mass is replaced with an 80-kilogram mass?
3. A mass weighing 24 pounds, attached to the end of a spring, stretches it 4 inches. Initially, the mass is released from rest from a point 3 inches above the equilibrium position. Find the equation of motion.
4. Determine the equation of motion if the mass in Problem 3 is initially released from the equilibrium position with an initial downward velocity of 2 ft/s.
5. A mass weighing 20 pounds stretches a spring 6 inches. The mass is initially released from rest from a point 6 inches below the equilibrium position.
 - (a) Find the position of the mass at the times $t = \pi/12, \pi/8, \pi/6, \pi/4$, and $9\pi/32$ s.
 - (b) What is the velocity of the mass when $t = 3\pi/16$ s? In which direction is the mass heading at this instant?
 - (c) At what times does the mass pass through the equilibrium position?



Sol: ① diff of SHM
 $\frac{d^2x}{dt^2} + \left(\frac{k}{m}\right)x = 0$

$W = mg$
 $m = \frac{W}{g}$
 $= \frac{4}{32}$
 $= \frac{1}{8} \text{ slug}$
 $k = 16 \text{ lb/ft}$
 $\downarrow$ Spring Constant

$\frac{d^2x}{dt^2} + \frac{16}{(1/8)}x = 0$
 $\frac{d^2x}{dt^2} + 128x = 0$
 $\omega^2 = 128$
 $\omega = \sqrt{128}$
 $\omega = 8\sqrt{2}$

$x(t) = C_1 \cos \omega t + C_2 \sin \omega t$
 $= C_1 \cos 8\sqrt{2}t + C_2 \sin 8\sqrt{2}t$
 $T = \frac{2\pi}{\omega} = \frac{2\pi}{8\sqrt{2}}$
 $= \frac{\pi\sqrt{2}}{8}$

Sol: ② $f = \frac{2}{\pi}$, $f = \frac{\omega}{2\pi}$, $\omega^2 = \frac{k}{m}$

$f^2 = \frac{\omega^2}{(2\pi)^2}$
 $(2\pi)^2 f^2 = \omega^2$
 $4\pi^2 f^2 = \frac{k}{m}$
 $4\pi^2 \cdot \frac{(2)^2}{\pi^2} = \frac{k}{20} \rightarrow ①$
 $k = 320 \text{ N/m}$

Sol: ③

Replace $m = 80 \text{ kg}$ in ①

$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{320}{80}} = \frac{1}{\pi} \text{ cycles/sec}$

